Diesel Engines
V 4000 R 40

Maintenance Schedule
M050531/02E

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept.
Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.
The Maintenance Schedule is based on the power output defined in the individual orders, under consideration of the operating parameters established in the order's specifications.
The time intervals according to which the maintenance tasks and the relevant checks and tasks involved are to be completed are based on operational experience and can only serve as guidelines. Specific operational conditions may require modification of the Maintenance Schedule.

Note:

The engine manufacturer cannot be held responsible for the performance of the following maintenance tasks. The operator is responsible for carrying out the following maintenance tasks:
Fuel filter
The maintenance interval depends on the degree of fuel contamination. The fuel filter must be replaced after 2 years at the latest.
Battery
Battery maintenance depends on the degree of use and ambient temperature. The battery has to be checked according to the manufacturer's instructions, as it is not always a part of the MTU scope of supply.
Fluids and Lubricants:
Only those fluids and lubricants are to be used that are approve of in the MTU Fluids and Lubricants Specification A001061. A special note must be made that, after maintenance or repair work on the cooling system, the operator must check the concentration of the corrosion inhibitor or the corrosion-inhibitor antifreeze.

Major Overhaul

Requires complete engine disassembly to facilitate repair/overhaul of all relevant components.

Out-of-Service Periods

If the engine is to remain out of service for several weeks, spray engine oil into the air-intake ducting. Actuate the engine shutdown system and bar the engine with the starting equipment when performing this task. Carry out maintenance tasks in accordance with the time limit schedule (see the Maintenance Frequency Chart). In addition, the engine must be barred manually every month. If the engine is to remain out of service for more than 1 month, carry out engine preservation in accordance MTU Fluids and Lubricants Specification, Publication No. A001061.

Application group
2A: Rail Traction

Maintenance Frequency Chart

Maintenance Level	W1:	Operational checks, daily	X
	W2:	Operating hours Limit, months	500 6
	W3:	Operating hours Limit, year	1 000 1
	W4:	Operating hours Limit, year(s)	5 000 2
	W5:	Operating hours Limit, year(s)	10 000 6
	W6:	Operating hours Limit, year(s)	30 000 18

Maintenance Task Schedule

Code No.	Maintenance Level W1 – Operational checks	
G000.000.13	Engine operation	Check speed
G000.000.17	Engine operation	Check temperatures (if gauges are provided)
G000.000.19	Engine operation	Check running noises
G000.000.21	Engine operation	Check engine and external pipework for leaks,
G000.003.15	Engine operation	Check pressures (if gauges are provided)
G083.101.03	Fuel pre-filter	Drain coolant and contaminants
G083.101.17	Fuel pre-filter	Check differential pressure
G101.011.05	Turbocharger connections	Check exhaust connections for leaks Check oil supply/return pipework for leaks
G121.052.01	Air filter	Check service indicator
G124.051.01	Intake air line	Check drain line for water discharge and obstructions
G140.000.05	Exhaust system	Check exhaust gas color
G140.000.11	Exhaust system	Drain condensate (if drain valve provided)
G180.000.01	Engine oil	Check level
G202.000.03	Engine coolant	Check level
G510.070.01	Monitoring system	Carry out lamp test

	Maintenance Levels	W 2 3 4 5 6
G180.000.04	Engine oil	Take sample and analyze
G180.000.05	Engine oil	Change – With oil category 1, with centrifugal oil filter Every 500 operating hours – With oil category 2, with centrifugal oil filter Every 1000 operating hours – With oil category 3, with centrifugal oil filter Every 1500 operating hours After 2 years at the latest
G183.052.03	Engine oil filter	Replace – Accomplish when changing the oil
G183.101.01	Centrifugal oil filter	Clean, check thickness of oil sludge layer Replace paper sleeve
G055.050.01	Valve gear	Check valve clearances
G083.051.03	Fuel duplex filter	Change throw-away filter
G083.101.09	Fuel pre-filter	Replace filter element and sealing ring
G083.101.15	Fuel pre-filter	Clean
G121.051.03	Air filter	Clean, empty dust collection box
G140.000.11	Exhaust system	Check, check drain for obstructions
G180.000.05	Engine oil	Change – With oil category 2, with centrifugal oil filter each W3 maintenance level = 1000 operating hours After 2 years at the latest
G183.052.03	Engine oil filter	Replace – Accomplish when changing the oil
G202.000.05	Engine coolant	Take sample and analyze – Change coolant, see MTU Fluids and Lubricants Specification
G202.051.01	Engine coolant pump	Check relief bore for obstructions
G203.000.07	Charge air coolant	Take sample and analyze, change if necessary (see MTU Fluids and Lubricants Specification A001061)
G203.051.01	Charge air coolant pump	Check relief bore for obstructions
G205.051.01	Engine coolant cooler	Check cooler elements for external contamination
G205.051.03	Charge-air coolant cooler	Check cooler elements for external contamination
G365.000.03	Exhaust system	Check, check drain for obstructions
G018.101.02	Crankcase ventilation	Replace filter element
G073.051.07	HP fuel delivery pump	Check relief bore for obstructions
G075.051.05	Fuel injector(s)	Replace
G121.051.01	Air filter	Replace filter elements
G213.051.01	Charging generator	Check condition of coupling
G231.051.01	Engine mounts	Check tightness of securing screws and nuts
G507.000.01	Engine control system	Wiring: Check security and condition
G510.000.01	Monitoring system	Check function of monitoring equipment
	Prior to commencing W5 maintenance, run the engine and record all measured values, then drain engine coolant and flush cooling system!	
	Valve gear	Remove and check rocker arms and valve bridges Check swing followers, rollers, ball sockets and swing follower bearings for wear
	Cylinder heads	Replace
	Running gear	Check piston crowns (visual inspection)
	Crankcase	Check cylinder liner wear pattern
	Vibration damper	Replace
	Turbocharger	Replace
	Air ducting	Remove, clean and replace seals
	Intercooler	Replace
	Starter	Replace
	Charging generator	Replace
	Exhaust silencer	Clean, replace seals
	HP fuel delivery pump	Replace
	HP fuel delivery-pump sensor	Replace
	Engine coolant pump	Replace
	Charge air coolant pump	Replace
	Engine coolant cooler	Clean and leak test
	Charge-air coolant cooler	Clean and leak test

	Engine coolant thermostat	Check, replace if necessary		
	Charge-air coolant thermostat	Check, replace if necessary		
	Engine coolant preheating	Overhaul		
	Engine oil heat exchanger	Remove, clean, leak test and replace seals		
	Centrifugal oil filter	Check, replace bearings if necessary		
	Major Overhaul	Total disassembly of the engine; inspection / repair of all components		